Case II: Stock-Out vs. Overstock

Fresh Bread Inventory at Save Supermarkets

P300: Team [#] | [Student Name 1] [Student Name 2] [Student Name 3] [Student Name 4]

Prepared by Annie Perkins, Inventory Manager, for Finance, Replenishment, Operations and Supply Chain

Executive Summary

Fresh bread is one of Save Supermarkets' five best sellers, yet the network throws away about 10.3% as many loaves as it sells. This report sets how much bread to order, when to order it, and what it costs to raise our service level from 80% to the managers' 90% goal and my long-term 95% goal.

The Economic Order Quantity (EOQ) is 12,852 loaves, for an annual total inventory cost (TIC) of $575,089. That order lasts about 6.6 days and fits in one 53-foot refrigerated trailer, but it is more than twice the bread's three-day shelf life. The reorder point is 9,784 loaves.

Under a daily (single-period) policy, safety stock is 225 loaves at 80%, 342 at 90% and 439 at 95%, giving annual TIC of $577,302, $577,143 and $577,024. Counting daily waste and lost sales, 90% costs about $6,800 more per year than today and 95% about $14,900 more.

I recommend moving to 90% now with daily orders, and reaching 95% by improving forecasts rather than by adding bread.

1. Introduction and Purpose

As Inventory Manager, I reviewed fresh bread replenishment across our 25 stores for the Finance, Replenishment, Operations and Supply Chain teams. Running out of bread costs us $0.78 in profit per loaf, while each unsold loaf costs $0.35 to dispose of. We currently have enough bread on about 80% of days (an 80% service level), and store managers want 90%. This report answers how much to order, when to order, and what each service level costs.

2. Methods and Analysis

Exhibit 1 gives 181 days of sales and disposal, which we treat as normally distributed loaves. Average daily demand (μ) is 1,956.67 loaves with a standard deviation (σ) of 266.78. Following the case, the carrying-cost percentage (Ci) is the average daily disposal rate, 10.34%. Other inputs: annual demand (D) of 700,000 loaves, unit cost (U) of $0.82, ordering cost (Co) of $10 and a five-day lead time. We used three tools:

- Economic Order Quantity (EOQ): the order size that balances ordering cost (high when we order often) against carrying cost (high when we order a lot). EOQ = $\sqrt{(2 \cdot D \cdot Co / (U \cdot Ci))}$. Total Inventory Cost: $TIC = Co \cdot (D/Q) + U \cdot Ci \cdot (Q/2) + D \cdot U$.
- Reorder Point (ROP): the stock level that triggers a new order. ROP = average daily demand × lead time, with no safety stock.
- Single-period model: for a product that spoils quickly, order expected demand plus a safety buffer: $Q = \mu + z\sigma$. Safety stock is $z\sigma$, where z is 0.842, 1.282 and 1.645 for 80%, 90% and 95% service. Each Q was then placed into the TIC equation.

3. Results

Q1. Trade-offs of carrying more or less bread

Table 1. Inventory trade-offs for fresh bread

Type	Reasons to carry more bread	Reasons to carry less bread
Financial	Avoids losing $0.78 profit per loaf short; fewer orders lower the $10 ordering cost; full trailers lower freight cost per loaf.	Each unsold loaf costs $0.82 plus $0.35 to dispose of; ties up cash; uses limited refrigerated and frozen space.
Non-financial	Higher service level and customer satisfaction; protects store traffic for a top-five product; covers demand spikes.	Fresher product and better quality image; less food waste, in line with our mission; less labor for rotation and disposal.

Q2 to Q5. Continuous review

Table 2. Continuous review results

Question and calculation	Result
Q2. EOQ = √(2 × 700,000 × $10 ÷ ($0.82 × 10.34%))	12,852 loaves
Q2. TIC = $544.67 ordering + $544.67 carrying + $574,000 purchase	$575,089.35 per year
Q3. Days of supply = 12,852 ÷ 1,956.67	About 6.6 days
Q4. Trailer capacity = 26 pallets × 500 loaves	13,000 loaves: yes, it fits (25.7 pallets)
Q5. ROP = 1,956.67 × 5 days	9,784 loaves

At the EOQ, ordering and carrying costs are equal ($544.67 each), and purchase cost makes up 99.8% of TIC. The EOQ is 25.7 pallets, which fits in one 26-pallet trailer (13,000 loaves). However, bread lasts only three days fresh and frozen storage holds just one pallet (500 loaves), so more than half of an EOQ delivery would spoil. The five-day lead time is also longer than the shelf life.

Q6 and Q7. Safety stock and cost by service level

Table 3. Safety stock, order quantity and annual TIC by service level

Service level	z-value	Safety stock (zσ)	Order quantity (μ + zσ)	Annual TIC
80% (current)	0.8416	225 loaves	2,182 loaves	$577,301.69
90% (managers)	1.2816	342 loaves	2,299 loaves	$577,142.81
95% (long term)	1.6449	439 loaves	2,396 loaves	$577,023.70

Each step up in service needs a larger buffer: 117 more loaves per day to reach 90%, and another 97 for 95%. Under the TIC equation, all three daily policies cost about $2,000 more than the EOQ, mainly from ordering about 320 times a year instead of 54. TIC falls slightly as service rises because larger daily orders mean fewer orders, while carrying cost rises from $92 to $102.

The TIC model charges carrying cost only on average inventory, so it understates daily waste. When we also count $0.35 disposal and $0.78 lost profit per loaf over 312 selling days, expected annual cost is $35,018 at 80%, $41,787 at 90% (+$6,769) and $49,883 at 95% (+$14,865).

4. Conclusions and Recommendations

The EOQ minimizes cost on paper but ignores spoilage, so bread should be managed with the single-period model. I recommend:

- Adopt a 90% service level now by ordering about 2,299 loaves per day (μ + 1.28σ). The extra cost of about $6,800 a year is roughly 1.2% of bread spend and keeps a top-five product on the shelf.

- Reach 95% by reducing demand variability rather than adding bread. Forecasting by day of week and by store would lower σ, and safety stock falls in direct proportion.
- Shorten the five-day lead time with daily supplier deliveries, or use par-baked or frozen bread with more freezer space.
- Recover value from leftover bread through day-old markdowns and food bank donations instead of paying $0.35 per loaf to dispose of it.
- Track in-stock rate, disposal rate and forecast error weekly, and update μ and σ monthly.